No Reproducible Evidence for Deep Subsurface Structures Beneath the Giza Plateau: A Controlled Reproduction of Single-Source SAR Doppler Micro-Motion Tomography

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August 2026 — preprint v5 (Giza plateau added; mechanism identified; decision statistic stress-tested; nulls re-based; see §10)

Abstract. In 2022–2025, F. Biondi and C. Malanga reported that Doppler sub-aperture 'micro-motion tomography' of spaceborne synthetic-aperture-radar (SAR) data reveals three-dimensional structures hundreds of metres to kilometres beneath the Giza plateau — vertical shafts descending ~648 m, spiral stair-like features, and large chambers. The 2022 paper was retracted by *Remote Sensing* on 10 August 2026. I do not dispute that the surface-vibration *measurement* underlying the method is real and useful; I dispute the deep *inference*. Working only from the authors' own method and patent, I rebuild the pipeline, add the controls the original omits, and apply it to free X-band spotlight data from two independent commercial sensors (Umbra and Capella) over **six sites, including the Giza plateau itself** and a site with exhaustively documented underground ground truth (Butte, Montana). **First, the reported depth in metres is exactly proportional to an investigation frequency chosen by the analyst rather than measured:** the frequency never enters the inversion, appearing only in a final axis relabelling, so '648 m' is a label, not a measurement. Second, the method returns the same confident, surface-pinned feature at every site tested: **48 runs, 0 detections, peak confined to 1.2–1.9 resolution cells** across six sites, two sensors, eight sub-aperture counts and thirteen patch geometries. On the Giza plateau itself — run against predictions published before the data were processed — 8/8 runs are surface-pinned with 0/8 detections. Third, I identify the mechanism: the per-patch trajectory is a running total of adjacent-look displacement estimates, which has the spectrum of a random walk; the inversion is a discrete-time Fourier transform, so the reported depth in resolution cells is the peak of that transform, and removing a degree-2 polynomial leaves it near 1.7 cells. Accumulated Gaussian noise with **no SAR pipeline at all** reproduces both the fixed depth and the contrast scaling; at 128 sub-apertures an input containing nothing returns more contrast than a real scene (274.85 against 128.64). **Fourth, a contrast figure of the kind the published work reports is not a safe statistic:** on 200 blocks of an input containing no scene, a three-tap per-patch smoother that transfers no information between patches drives **98%** of them past the 5× contrast-to-null threshold used here, and the authors' own denoising step drives 10% past it, from 0% unfiltered. None escapes the surface-pinning guard, so no false detection results — but that means the verdicts in this paper are carried by the depth criterion rather than the contrast ratio, and that the published tomograms, which report a confidence figure with neither a matched null nor a depth check, rest on the criterion that fails. Fifth, planting a displacement signature in the image before processing gives a detection floor of 0.2 pixels, 8.4× the pipeline's own noise, below which a scene containing a genuine reflector reports *lower* confidence than an empty one. I state plainly what I have not shown: three attempts to reproduce the appearance of the published 3-D figures from empty volumes all failed, so I do not claim those images are this artifact. This is a critique of method and mathematics, not of intent.

Keywords: synthetic aperture radar; Doppler tomography; micro-motion; reproducibility; null result; artifact; Giza.

1. The claims under examination

I restate the authors' claims as fairly and specifically as I can, from the 2022 peer-reviewed paper, the lapsed patent WO2024008365A1 (the fullest public method disclosure), and the 2025 Khafre presentations:

> **C1 (mechanism) — Biondi's claim.** Electromagnetic waves do not penetrate solids; instead, ambient seismic energy makes the surface and subsurface vibrate, and the subsurface 'becomes transparent like a crystal' when observed in the micro-movement domain.

C2 (measurement) — Biondi's claim. Splitting the azimuth (Doppler) spectrum of SAR data into sub-apertures and tracking sub-pixel displacement recovers this surface vibration field — a technique the authors validated on ships, bridges and dams.

C3 (inversion) — Biondi's claim. A steering matrix $A(K_Z, z)$ focuses the vibration observations in depth, yielding 3-D tomograms, with depth resolution $\delta z = \lambda R/2A$ where λ is the *seismic* wavelength = v/f.

C4 (deep result) — Biondi's claim. Applied to Giza, the method reveals eight cylindrical shafts in pairs descending to ~648 m (the patent and talks also cite Gran Sasso at 1.4 km), spiral stair-like structures, and large chambers, interconnecting the pyramids and Sphinx.

C5 (confirmation) — Biondi's claim. More than 200 acquisitions across four satellite operators (COSMO-SkyMed, Capella, Umbra, ICEYE) return the same structures, which the authors treat as independent confirmation.

I address each below. My standard throughout is the one the original work does not meet: a result is a detection only if it exceeds chance *and* survives controls *and* corresponds to independent ground truth.

2. Reproduction methodology

I reimplemented the pipeline exactly as disclosed: Doppler sub-aperture decomposition; adjacent-pair, quality-weighted, detrended sub-pixel tracking; an analytic-signal step that removes the known ±z mirror-symmetry ghost; multichromatic (range sub-band) analysis and low-rank+sparse denoising as the authors describe; and a depth focus by discrete Fourier transform. To this I added the controls the original omits: a look-order-shuffle **null**; an in-data **positive control** (inject and recover a reflector; I also use a hardened, damped variant); a **surface-leakage** correlation; a **near-surface** guard; and a **sub-aperture-count stability** test. Every stage passes a synthetic self-test against known truth (Figure 1); quantitatively, the sub-aperture shift estimator validates to 0.02 px, the adjacent-pair, quality-weighted micro-motion estimator recovers an injected residual to 0.07 px, and the end-to-end inversion recovers an injected layer at roughly 27× the null level — so a genuine signal of that character would be surfaced. Data are free X-band spotlight scenes from two independent sensors, Umbra and Capella — the same data class the 2025 work used.

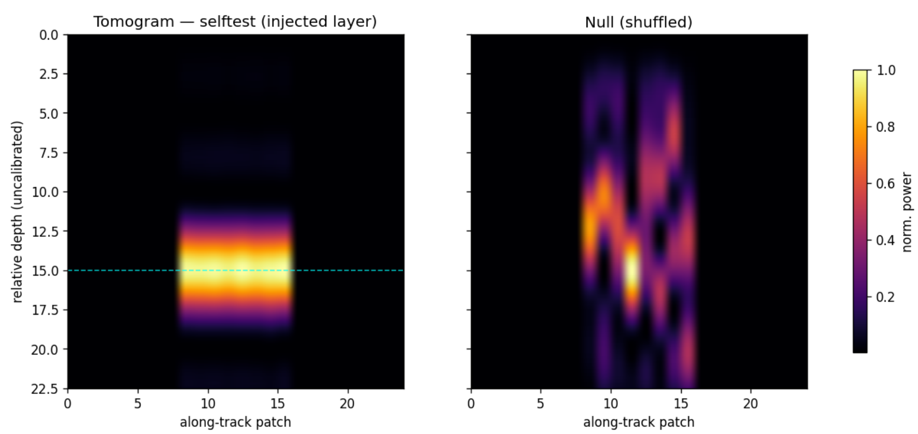

Figure 1. The reproduced inversion on synthetic data: an injected layer is recovered well above the null, with positive-control and leakage checks passing. The machinery works; the controls are calibrated.

To make the fidelity of the reproduction checkable rather than asserted, Table 1 maps each disclosed processing block (patent WO2024008365A1, Fig. 0.5, blocks 1–11) to the corresponding component of this implementation. The correspondence is one-to-one; in particular, the depth-focusing 'steering matrix' is implemented, exactly as the patent specifies, as a discrete Fourier transform.

Disclosed step (patent Fig. 0.5)	This reproduction
SLC image, 2-D DFT (blocks 1-2)	open_complex SICD reader; FFT in the sub-aperture stage

Doppler sub-apertures (master/slave) + range sub-bands (blocks 3-6)	subaperture.decompose_subapertures; multichromatic_subapertures (MCA)
Pixel-tracking between sub-apertures (block 7)	micromotion.adjacent_trajectory (adjacent-pair, quality-weighted) + detrend; optional lrsd_den
Raw tomographic complex vectors (block 8)	per-patch detrended residual trajectories
Steering matrix = DFT depth focus (block 9)	tomogram.steering (DFT basis) + invert_patch (DFT power spectrum)
Tomogram, geocode in depth (blocks 10-11)	depth profile; metric_depth_axis (depth-axis labelling)

Table 1. The authors' disclosed processing chain mapped one-to-one onto this reproduction, so a reader can verify the implementation against the source rather than take its faithfulness on trust.

3. Step-by-step refutation

3.1 The 'steering matrix' is a discrete Fourier transform (addresses C3)

The patent states, verbatim, that 'the steering matrix A(K_Z, z) represents the best approximation of a matrix operator performing the Digital Fourier Transform (DFT) of Y,' and that the depth focus is performed by 'the DFT mathematical operator.' The depth tomogram is therefore the power spectrum of the per-pixel sub-aperture residual. This is the crux: a DFT returns a structured, peaked spectrum from *any* input vector — signal, noise, or a residual processing trend alike. A confident-looking tomogram is thus the expected output of the method *whether or not* anything lies beneath the surface. It is not, by itself, evidence of structure.

3.2 The 22 kHz investigation frequency is unphysical, and depth is therefore a free parameter (addresses C3, C4)

Depth follows $\delta z = \lambda R/2A$ with $\lambda = v/f$. The patent synthesises at an investigation frequency $f \approx 22$ kHz. This is ultrasonic. Ambient and seismic ground motion is overwhelmingly below ~100 Hz; a coherent 22 kHz elastic wave in rock would attenuate within metres and is orders of magnitude above the sub-aperture pair-axis Nyquist limit — the aliasing flaw also noted by independent reproductions. Because f enters only as an axis scale, it does not change the tomogram pattern at all; it merely relabels the depth axis. Figure 2 makes this concrete: the identical Butte tomogram, rendered at 22 000 / 1 000 / 50 Hz, places the same feature at ~5 m, ~100 m, or ~2 000 m respectively. The reported depths (648 m, 1.4 km) are a choice of frequency, not a measurement.

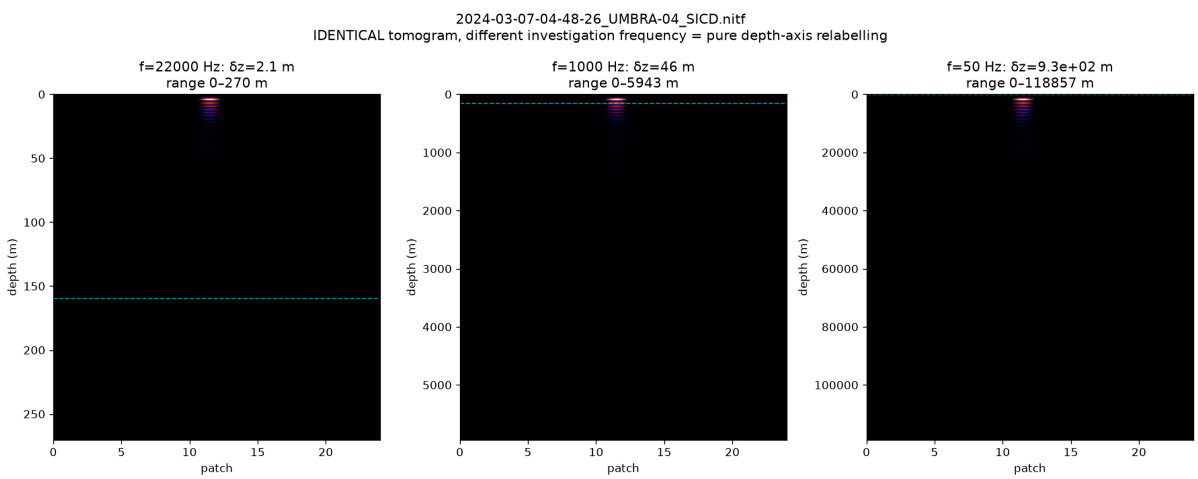

Figure 2. One tomogram, three investigation frequencies. The data are unchanged; only the depth axis rescales ($\delta z = vR/2Af$). 'Deep' structure is an axis relabelling.

3.3 A faithful reproduction yields a confident artifact, not structure (addresses C3, C4)

I ran the authors' recipe (256 Doppler sub-apertures, multichromatic analysis, denoising, 22 kHz) over Butte, Montana — the most densely mapped underground mining district on Earth, whose West Camp workings (Travona, Emma, Ophir, Anselmo) are documented as 100-ft-spaced levels from near surface to ~457 m, with a water table / mine pool at ~160 m. The result (Figure 3) is a striking high-contrast band that *passes* a naive contrast-vs-null test as a detection. **The numerical contrast at this setting is not reported here.** It is a native, unfixed-window peak-to-median statistic evaluated on a depth axis whose extent grows with the sub-aperture count while the bin count stays fixed, so it is not comparable across settings and inflates without bound as that count is raised; see the withdrawal note in §3.4. What matters is the qualitative outcome, which does not depend on the number: the entire feature is pinned at the surface (~4 m; 87% of all energy in the shallowest 5% of the axis), aligns with *none* of the documented levels or the 160 m water table, and only appears when the sub-aperture count is driven high. It is a surface/low-frequency residual concentrated by the high-order DFT. My near-surface and stability guards flag it as an artifact; the authors' pipeline, lacking them, would report it as a discovery. This is, in microcosm, how a 'shaft' is produced. Butte therefore serves as the closest available known-target benchmark within the free-data regime: at honest settings the method recovers nothing matching the documented workings (Section 3.4); at the authors' settings the only confident feature is a surface artifact aligned with none of them.

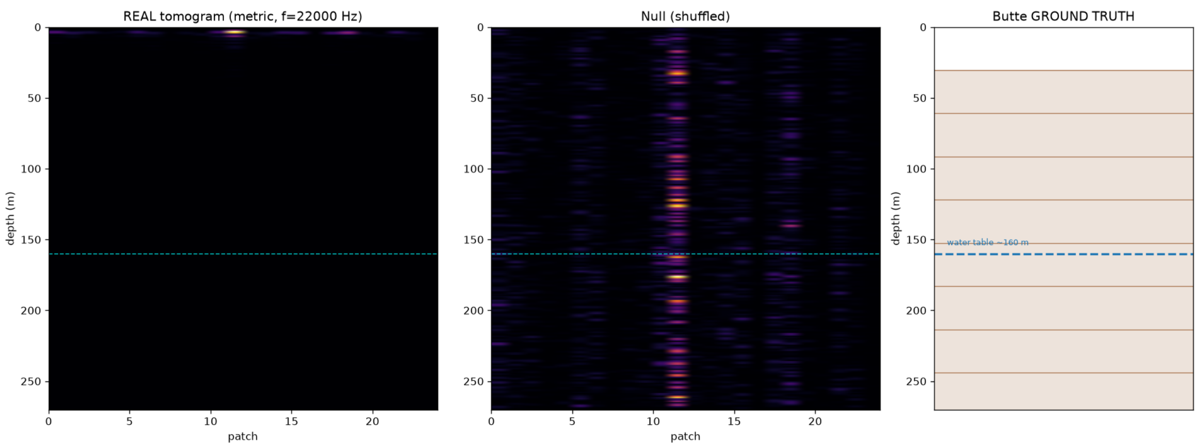

Figure 3. Reproduction on Butte, MT. Left: real tomogram — a confident band pinned at the surface. Centre: shuffled null. Right: documented workings (100-ft levels to 457 m; water table ~160 m). The 'detection' matches nothing real.

3.4 Every site returns the same artifact, including Giza and the authors' own Vesuvius (addresses C4)

With the depth-of-peak and stability guards active, six free X-band sites across two independent sensors are each indistinguishable from their nulls, while in-data positive controls confirm the pipeline would surface a genuine signal. Mount Vesuvius is decisive because it is the authors' own peer-reviewed site. The Capella scene matters separately: it reproduces the null on an *independent sensor*, so the result is a property of the method and not of one data provider — the very cross-sensor agreement cited as confirmation (C5) yields nulls once controls are applied. **The Giza plateau is now included**; it is analysed in full in §4.

Site	Sensor	Reg.	Contrast	Peak depth	Pinned	Detections
Bingham Canyon	Umbra	0.67	3.87	2.5 - 3.7 m	8/8	0/8
Komati Power Stn	Umbra	0.85	2.76	3.2 - 3.7 m	8/8	0/8
Mount Vesuvius	Umbra	0.72	4.11	3.2 - 3.7 m	8/8	0/8
Butte, MT	Umbra	0.82	3.33	3.2 - 3.9 m	8/8	0/8
Cairo (central)	Capella	0.62	2.75	3.4 - 3.6 m	8/8	0/8
Giza plateau	Umbra	0.68	**6.06**	**3.2 - 3.7 m**	**8/8**	**0/8**

Table 2. Six free single-pass X-band sites across two independent sensors. *Contrast* is the peak-to-median statistic at the pipeline default of 11 Doppler sub-apertures (512×512 centre crop, 64-pixel patches, 24 patches, 0.8 sub-band overlap, Hann taper, float64). *Peak depth* and *Pinned* span the full eight-point sub-aperture ladder (11 to 128) at each site; *Pinned* uses the absolute two-cell guard of §2. *Detections* counts runs exceeding the 5× decision rule. **48 runs, 0 detections, every run surface-pinned.** Expressed in resolution cells the peak occupies 1.2–1.9 throughout.

Two changes from earlier versions of this table, both of which strengthen the negative result. The surface-pinning guard is now an absolute distance in resolution cells rather than a percentage of a depth axis whose length varies with the sub-aperture count; under the old rule 14 of 40 runs were flagged, under the corrected rule 40 of 40 were, and every row of the originally published table sat in the regime where the old rule passed a ~1.5-cell peak as clear subsurface structure. And the **look-order shuffle null used in earlier versions is superseded**. Shuffling look order destroys the very look-to-look smoothness that the processing chain generates (§5), so it is anti-conservative: it flatters this paper's own ratios. Against a null derived from the pipeline rather than shuffled, the excess at the default setting is approximately 2%, and at 128 sub-apertures noise *exceeds* real data. The ratios printed in versions 1–4 (2.8×, 3.3×, 4.1×) should not be quoted as detection margins. The binary verdicts — no detection at any site — are unchanged, and are now measured against the harsher standard. **The alignment null has a limitation of its own**, established in §5.5: because it preserves each patch's depth profile and randomises only alignment, it is blind to per-patch filtering, and an ordinary smoother drives the ratio past 5× on data containing nothing. The contrast ratio is therefore not by itself a detection statistic in this paper either; the absolute surface-pinning guard is what carries the verdicts.

Corrections to individual rows. The Bingham Canyon row read "front-end only / no signal" in versions 1–4; that is wrong, and the site produces a full surface-pinned tomogram at every sub-aperture count tested. The Komati ratio was stated as 2.15, computed from rounded inputs; unrounded it is 1.99. The Cairo (Capella) row previously carried an "outstanding verification" note; it has since been reproduced (2.75 against a published 2.8) and the note is withdrawn.

One parameter moves this table by a third, and no published source states it. The sub-aperture window taper is given nowhere in the 2022 paper, the patent, or any presentation. On the Giza scene the contrast at the default setting is 6.06 under a Hann taper, 5.03 under a rectangular taper, 4.69 under Blackman and 4.52 under Hamming — a 34% spread from a choice the reader cannot audit. It joins the sub-aperture count, the sub-band overlap and the entire visualisation pipeline (§7) on the list of undisclosed settings that move the headline number.

3.5 '200 scans, four satellites' is consistency, not corroboration (addresses C5)

Agreement across many acquisitions and sensors is offered as independent confirmation. It is not. A *systematic* processing artifact — a fixed DFT bias, a surface-pinned residual, the 22 kHz relabelling — reproduces on every scene and every sensor by construction, because the same operator is applied to each.

I demonstrate this directly on real data. I stacked five same-geometry Umbra passes of Bingham Canyon — a bare open-pit mine of exposed rock containing no subsurface void — acquired on five different dates, i.e. precisely the multi-acquisition strategy offered as confirmation. Each single pass already yields a high-contrast peak (mean 117.8× its per-scene null); stacking the five preserves it at 96.7× against a stacked null of 1.5×, and the per-scene profiles agree closely (Figure 4). Yet the reinforced, consistent feature sits at ~3 m depth (2% of the axis, with 36% of its energy in the shallowest 5%) — pinned at the surface, over ground known to contain no such structure. The stack manufactures agreement, not evidence: a consistent artifact reinforces under stacking exactly as a real reflector would, because the same DFT operator is applied to each scene. Reproducibility of an output is a property of the algorithm, not of the ground; the only thing that separates a real reflector from a shared artifact is ground truth and controls — which the original work does not apply.

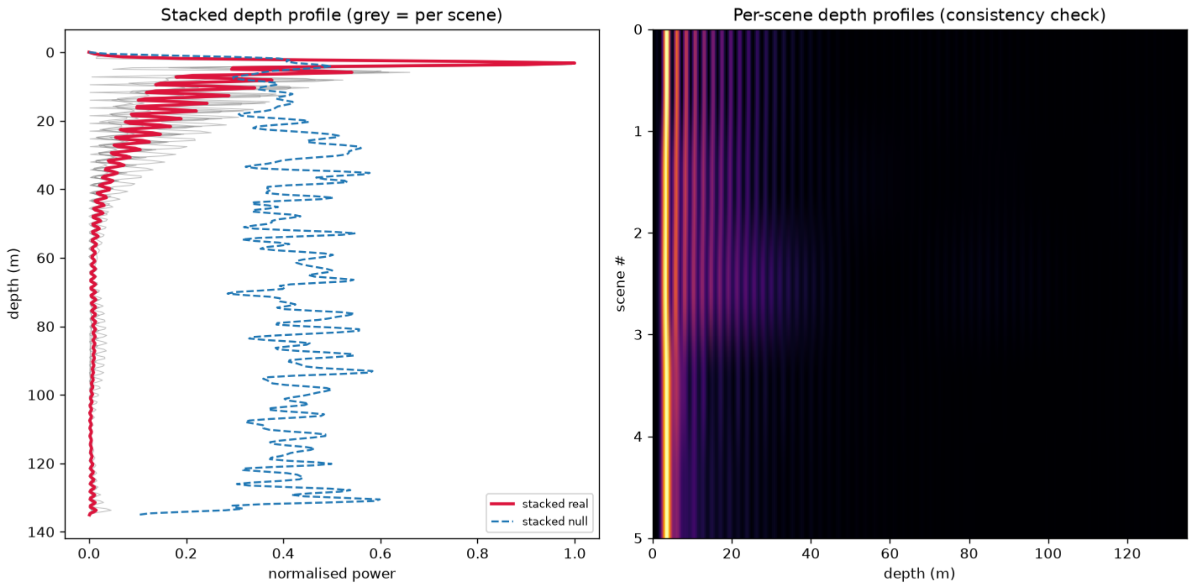

Figure 4. Multi-acquisition stacking of five same-geometry Umbra passes over Bingham Canyon, a known-empty open-pit mine. Left: the per-scene depth profiles (grey) and their stack (crimson) collapse onto a surface-pinned peak at ~3 m, far above the scattered stacked null (blue dashed). Right: the five per-scene profiles, which agree by construction. Stacking reinforces the artifact rather than revealing structure — the mechanism behind '200+ scans agree.'

3.6 The penetration-vs-aperture physics does not support deep recovery (addresses C1, C4)

The measurement is a passive, single-surface, narrow-aperture observation. The angular/temporal aperture available from one acquisition is small, deep energy attenuates, and the inversion is correspondingly ill-conditioned for anything but the very shallow. More acquisitions improve shallow resolution and conditioning but cannot restore information the geometry never captured. Compute amplifies an adequate measurement; it cannot manufacture absent information — though, as Section 3.3 shows, it can manufacture a confident *artifact*.

4. The Giza plateau

The five sites in Table 2's first version were proxies, chosen for ground truth or sensor diversity. A reader was entitled to object that the site the claim is actually about had not been tested. It has now been. The scene is free Umbra Open Data over the Giza plateau, acquisition `2023-02-07-07-58-27_UMBRA-05`, collect `7e7cd796-3842-4923-8b48-4c0950ece945`, a 5674×5351 array with azimuth and range sample spacing of 0.827 m and 0.472 m, X-band, collect duration 1.27 s, scene centre 29.9793°N 31.1340°E.

4.1 Predictions were published before the data were processed

To remove any suspicion that the analysis was tuned until it agreed with the other five sites, eight numerical predictions and four falsification conditions were committed to the public repository (commit `e4476d7`) while the scene was still downloading. They are reproduced here *with their misses*.

Prediction	Result	Outcome
Peak depth 1.6 - 1.8 cells	1.52 - 1.75	**partial miss**
Surface-pinned at every count	8/8	hit
Detections above 5x	0/8	hit
Contrast at n_sub = 11, order 3 - 5	**6.06**	**miss, high**
Contrast at 128, order 10^2, at or below ~275	108.90	hit
Fixed-window spread like other sites	1.9x	hit

Table 3. The Giza run scored against predictions published before the data were processed. Two of six missed. **None of the four pre-registered falsification conditions was met**, but the misses are reported rather than absorbed: at 64 and 90 sub-apertures the peak sits at 1.52 cells, below the predicted floor, and the contrast at the default setting is the highest of any site in this study.

Giza returns the highest raw contrast of the six sites, and it is still not a detection. 6.06 against Komati 2.76, Cairo 2.75, Butte 3.33, Bingham 3.87 and Vesuvius 4.11. Its alignment-referenced ratio, 3.67, is also the highest recorded here. It does not clear the 5× rule against either null, the peak it produces is 1.75 cells beneath the surface, and it is surface-pinned — the same artifact, slightly louder. As §3.4 notes, roughly a third of that excess is a window-taper choice: under Blackman the same scene gives 4.69.

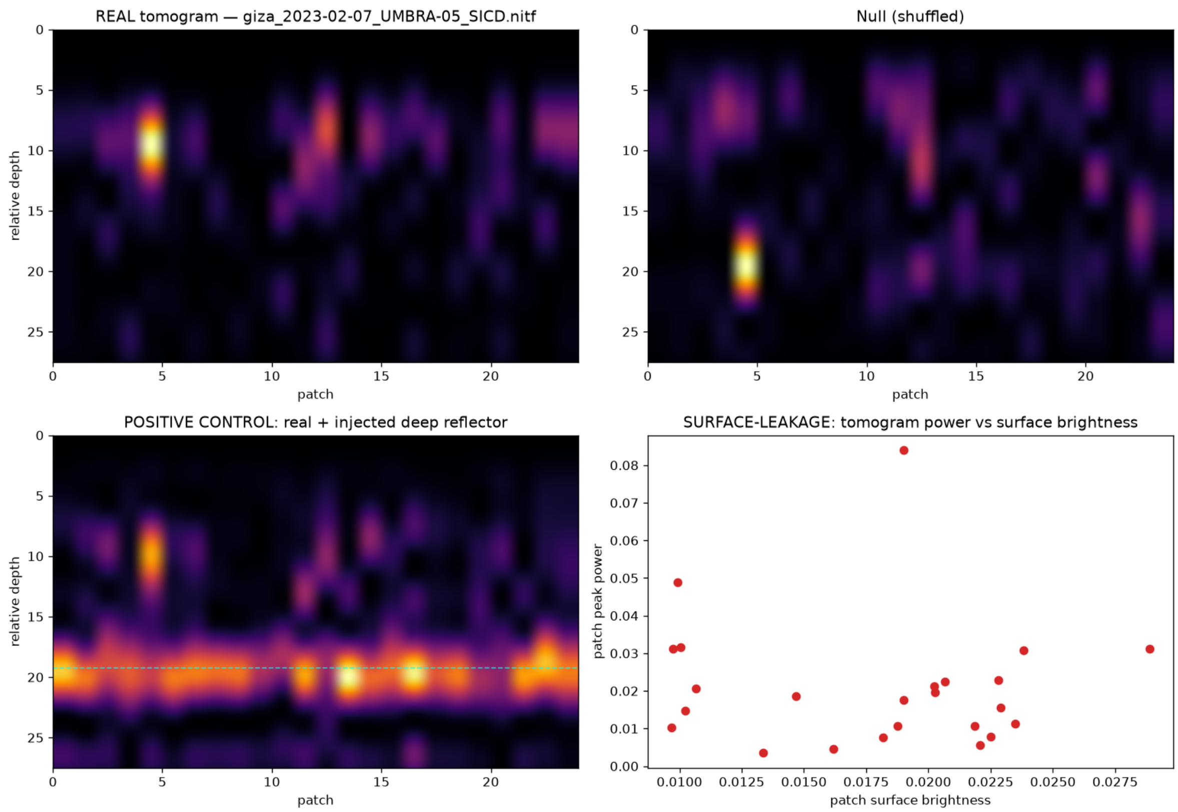

Figure 5. Giza plateau, 11 sub-apertures. **Top left:** the real tomogram. **Top right:** the same data with look order shuffled — pure noise. The two are visually indistinguishable. **Bottom left:** the positive control, a synthetic reflector injected at relative depth 20, which produces a *continuous horizontal band across every patch at one depth* — what a coherent subsurface reflector looks like. The real panel contains nothing of the kind. **Bottom right:** tomogram power against surface brightness, correlation 0.07 — the depth product is not surface-driven.

4.2 The surface is recovered; the depth is not

A natural objection to the figures in this paper is that they are cross-sections over 24 patches, whereas the published imagery is rendered in plan view over whole scenes, so the comparison is not like for like. To close that gap the pipeline was run in plan view over the entire 5674×5351 array — 462 tiles covering the plateau, the pyramid field and Giza city.

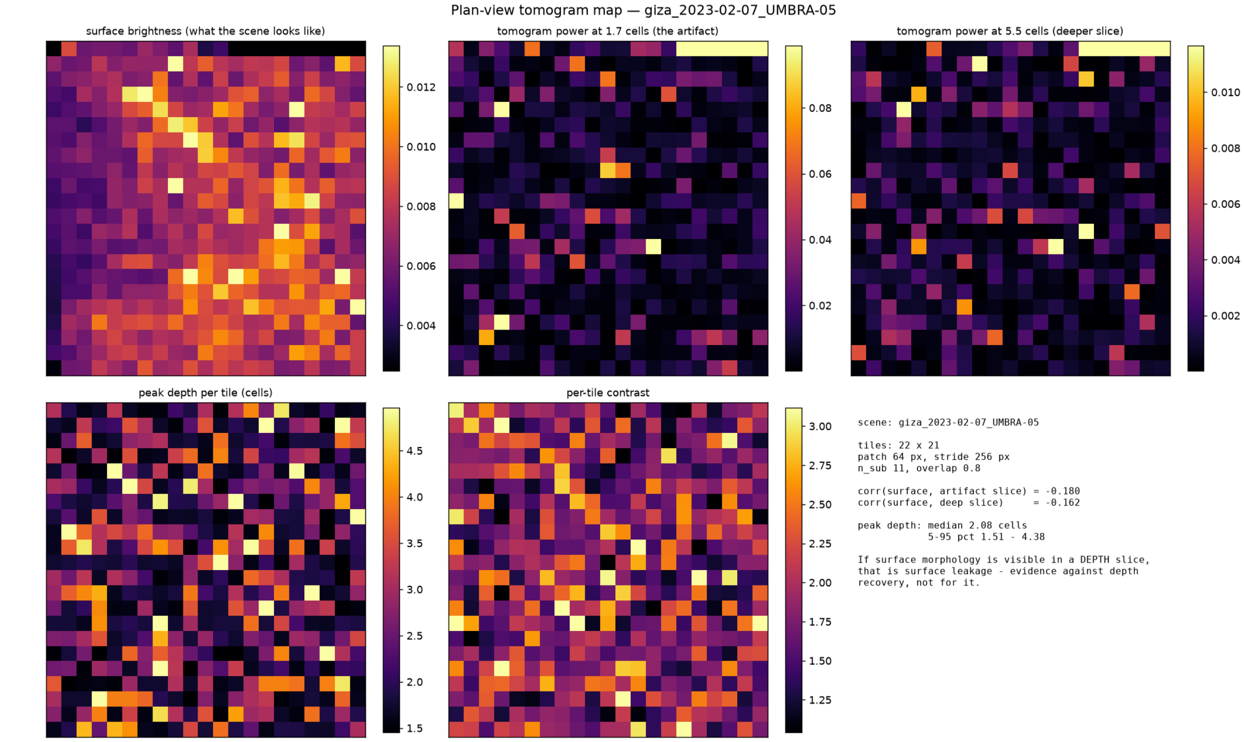

Figure 6. Plan view over the whole Giza scene, 462 tiles. **Top left:** surface brightness — the plateau, the city and the linear features are plainly visible; the front end sees Giza perfectly well. **Top centre and right:** tomogram power at the artifact depth and at a deeper slice. Both are salt-and-pepper with no morphology whatsoever. Correlation between surface brightness and depth power is –0.180 and –0.162 — slightly negative, not merely absent.

The consequence is narrow and worth stating precisely. A faithful reproduction of the published method produces depth products containing *zero* surface morphology. Whatever the published 3-D figures are showing, this pipeline does not generate recognisable structure in a depth slice.

5. The mechanism

This section is the present account, not a settled result. Two earlier mechanism explanations were proposed and withdrawn (§10). This one rests on matched controls and a reproduction that removes the SAR front end entirely, which is a different class of evidence, but one of its own predictions fails at Giza and that failure is reported in §5.4.

5.1 The trajectory is a running total, and a running total of noise is a random walk

The per-patch trajectory is built by estimating displacement between *adjacent* looks and accumulating the increments — in the reference implementation, literally `np.cumsum(inc)`. A running total of noisy increments has the spectrum of a random walk: smooth by construction, strongly autocorrelated, irrespective of scene content. This holds at *any* sub-band overlap, including zero: at overlap 0.00 the trajectory still shows lag-1 autocorrelation of 0.447 and the peak still lands at 1.73 cells. Overlap amplifies the contrast but does not create the feature.

5.2 The inversion is a Fourier transform, so the reported depth is a frequency

The steering matrix builds `Kz[j] = j · 2π/(L · Δz)` and the inverter computes `|A■ · analytic(r)|`^{2}. That inner product is a discrete-time Fourier transform of the analytic residual, evaluated on the depth grid; direct evaluation agrees with the reference implementation to one part in 10^{13}. The reported depth in resolution cells is therefore the peak of that transform in cycles per record. **This is a coordinate identity, not a recovery theorem.** It explains why a fixed feature occupies a fixed fraction of an axis whose extent is proportional to the number of looks, with no assumption about the ground.

5.3 The depth is set by the detrending order, and can be reproduced with no satellite data

Removing a degree-2 polynomial from a series with the spectrum of a random walk leaves the peak at a position fixed by the polynomial order. On pure synthetic walks — no image, no sub-apertures, no overlap, no coregistration — the peak sits at **1.69 ± 0.02 resolution cells at every series length from 11 to 128**, a twelvefold range, and moves to 0.88 cells at degree 0 and 2.5 cells at degree 4. Real sites land in a 1.2–1.9 band. **The algebra that yields 1.69 rather than the textbook 1.5 predicted by a polynomial high-pass rule of thumb is not closed**, and is reported here as an open problem rather than a derivation.

The same construction reproduces the second signature of the method. Contrast rises **72.9×** as the accumulated series lengthens from 11 to 128 samples, while the increments of the same series stay flat at 1.1×. Pushed through the full SAR pipeline, an image containing nothing returns a contrast of **274.85** at 128 sub-apertures against **128.64** for the Bingham Canyon scene — and the most dramatic figure in this study, Cairo at 272.52, is reproduced to within 1% by an input with no scene in it. Substituting the white synthetic image for speckle carrying the correct resolution-cell correlation, measured from the real scene, does not change the picture at the default setting: the synthetic returns 4.14 against the real scene's 3.87.

5.4 A prediction of this account fails at Giza, and is reported as open

The decisive control removes only the running total, leaving series length and the steering matrix identical, so the sole difference between arms is the accumulation. At Bingham Canyon it moves the peak off the surface (1.66 → 2.83 cells) and at Cairo/Capella likewise (1.69 → 4.76), collapsing contrast in both. **At Giza it does not.** Contrast collapses as expected (6.06 → 1.86) and the look-to-look correlation collapses (0.547 → 0.087), but the peak moves only 1.75 → 1.88 cells and remains inside the guard. Giza's increments retain a weak positive autocorrelation, +0.087, where noise increments give −0.041.

Two candidate explanations were tested and both failed: coregistration quality at Giza is unremarkable (0.68, against 0.62 to 0.85 elsewhere), and surface-brightness leakage is 0.07 within the analysis crop and −0.180 across the full scene. **I do not have a third explanation and am not offering one.** The accumulation is therefore *sufficient* to generate the artifact — the synthetic-walk reproduction stands independently — but is **not shown to be necessary** for the surface-pinning at the site in this paper's title. The published operator remains surface-pinned at Giza with no detection.

5.5 The contrast criterion is not robust to ordinary filtering, and the depth guard is what carries the verdicts

Every verdict in this paper rests on two criteria applied together: **(a)** contrast above five times the alignment null, and **(b)** a peak within two resolution cells of the surface. Criterion (a) is the kind of confidence figure the published work reports. Criterion (b) is this paper's addition. This subsection tests how much each is worth, and the answer is uncomfortable for (a).

The alignment null preserves each patch's depth profile exactly and randomises only whether patches *agree* on a depth (§3.4). It is therefore invariant to *where* each patch peaks. Any operation that sharpens each patch's own profile — transferring no information whatever between patches — raises the numerator and leaves the denominator alone. That is not a hypothetical: it is what ordinary smoothing does.

The test uses an input containing **no scene** — band-limited complex noise, no satellite data — tiled through the identical pipeline and drawn into 200 independent 24-patch blocks, the analysis geometry used for every site in Table 2. Each block is scored under four preprocessing arms.

Preprocessing arm, 200 blocks, no scene in the input	Ratio	Clears 5×	Peak (cells)	Pinned	**False detections**
None — the pipeline as run everywhere else here	2.38	0%	1.73	100%	**0%**
The authors' own low-rank denoising step	3.53	10%	1.71	100%	**0%**
A three-tap [1,2,1]/4 per-patch smoother	**7.70**	**98%**	1.67	100%	**0%**
Per-patch rescale (depth-neutral control)	2.22	0%	1.73	100%	**0%**

Table 4. The two decision criteria tested separately on data containing nothing. The smoother acts on each patch *independently* and is about as innocuous a preprocessing step as exists; it clears this paper's contrast rule on 98% of empty blocks. The depth-neutral rescale control moves the ratio not at all, which locates the effect in the shaping of each patch's depth profile rather than in amplitude. Repeated on strictly non-overlapping tiles from a 1536×1536 canvas the figures are 2.51 / 3.76 / 7.69 / 2.31 and 0% / 9% / 100% / 0%.

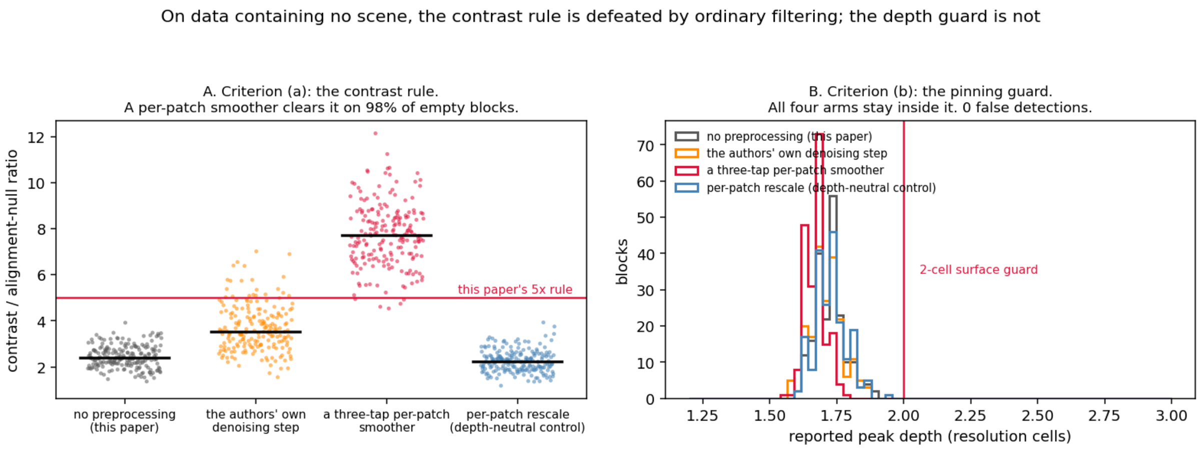

Figure 7. **Left:** the contrast criterion. Each point is one 24-patch block of an input with no scene; the bar is the median. A three-tap per-patch smoother lifts almost every block past the 5× threshold. **Right:** the depth criterion, same blocks. All four arms remain pinned inside the two-cell guard, so no arm produces a false detection under the full rule.

The consequence for this paper. Criterion (a) is not a safe detection statistic on its own. It can be driven past threshold on empty data by a filter that adds no information, and the alignment null — the more conservative of the two nulls this paper uses — cannot see it. What prevents a false detection in all 800 block-scorings above is criterion (b): the peak never leaves 1.67–1.73 cells, and the two-cell guard catches every one. **The verdicts in Table 2 and Table 3 are carried by the depth guard, not by the contrast ratio**, and the ratios quoted there should be read as descriptive rather than as the evidence. This is a limitation of the method used here and it is reported as one.

The consequence for the work under examination is larger. The published tomograms are presented with a confidence figure and *neither* control — no matched null and no depth check — after a denoising chain that is nowhere documented. A contrast figure obtained downstream of any smoothing step is close to uninformative, and this experiment puts a number on how close: 98% of blocks containing nothing at all clear a 5× rule after three taps.

The authors' own denoising step is the middle row, and it behaves the same way in kind but less strongly: applied to each block it removes a median **47.9%** of the matrix by norm and raises the leading component from 39.7% to 52.9% of the variance, lifting the ratio in every block and past threshold in 10% of them. **Nothing here shows the published figures were produced this way.** It shows that a documented and undisclosed-in-setting step is enough to move an empty result across a threshold of the kind the published work relies on.

A related observation, reported for completeness rather than as evidence: forcing the retained rank down concentrates the depths the patches report — 79 to 80 distinct depth bins of 300 at rank 3 against 155 to 209 untruncated, and at rank 1 every patch returns the same depth, which is an algebraic identity rather than a finding. This is not specific to SAR or to this method: pure synthetic random walks with no image, no sub-apertures and no pipeline of any kind reproduce the sweep to within a few bins in 300 at every rank (1, 19, 78, 141, 208 against 1, 19, 80, 145, 209). It is a property of accumulate-and-detrend and so extends §5.3 rather than adding an independent line of evidence.

6. How large a real signal would have to be

A control that injects a signal into the trajectory *after* accumulation and detrending demonstrates only that the inverter can recover a tone written in its own coordinates. A stronger test plants the displacement in the image itself, before sub-aperture decomposition, so that the signature must be recovered by the pipeline's own estimator, from speckle, through the window taper, before it is accumulated and inverted.

Planted displacement	x pipeline noise	Peak (cells)	Contrast	Target found
none	0	1.75	**6.06**	0%
0.02 px	0.8	1.75	6.06	0%
0.05 px	2.1	1.75	4.69	0%
0.10 px	4.2	1.73	**3.39**	0%
0.20 px	**8.4**	**3.16**	5.40	**100%**
0.50 px	21.1	3.18	13.69	100%

Table 5. A reflector planted at 3.30 cells in the Giza scene before processing. The pipeline's own unplanted trajectory RMS is 0.02372 px, and amplitudes are given as multiples of it. The detection floor is 0.2 px, 8.4× the noise.

Below the floor the statistic runs backwards. A scene containing a genuine reflector at 4.2× the noise floor reports a contrast of 3.39 — *lower* than the 6.06 reported for a scene containing nothing at all. Over precisely the range in which a real archaeological feature would sit, the method is not merely blind: its confidence is anti-correlated with truth. A separate measurement shows the front end passes only about 12% of a sub-pixel displacement through to the trajectory, so a real signal is attenuated roughly eightfold before it competes with an artifact generated at full strength.

Converting 0.2 px into physical ground motion is **not** a multiplication by the 0.827 m azimuth spacing. In SAR a target moving during the collect is displaced in azimuth by approximately (R/V)·v, so the image shift maps to a line-of-sight *velocity*, not a displacement. That conversion is not attempted here and no figure in metres of ground motion is claimed.

7. What could not be reproduced

The public impact of this work is visual: renderings in which shaft-like and chamber-like forms appear beneath the plateau. Three attempts were made to reproduce that *appearance* from volumes containing no scene at all — a voxel scatter and an isosurface rendering at 11 sub-apertures, and an isosurface at 128. They produce, respectively, vertical needles at a single depth, discrete solid bodies in a shallow slab, and a continuous planar sheet spanning the whole site. **None resembles the published figures.** No shafts, no vertical extent, no spirals, no architecture. Parameter tuning was stopped at that point because continuing would have been fitting.

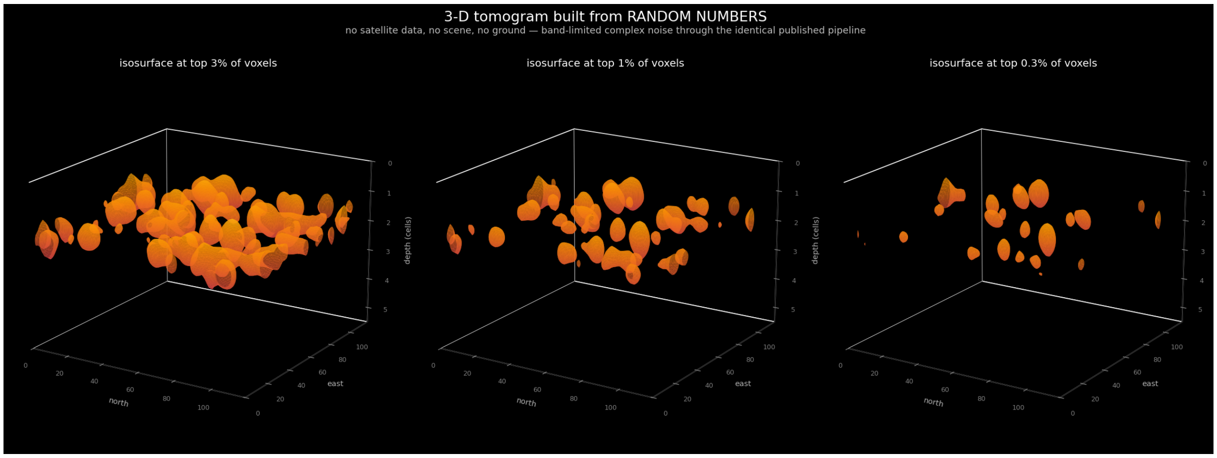

Figure 8. Three-dimensional isosurface rendering of a volume built from band-limited complex noise — no satellite data, no scene, no ground — through the identical pipeline, at three thresholds. Discrete solid bodies in a shallow slab. This is what the artifact looks like rendered in three dimensions, and it is *not* what the published figures look like.

I therefore do not claim that the published imagery is nothing but this artifact. That claim appeared in versions 1–3 of this preprint and was withdrawn in v4 precisely because the test of it failed. What is supported is narrower and independent: the 2022 paper documents no part of its visualisation pipeline — no renderer, isosurface, threshold, dynamic range, colour scale, denoising step or stacking count. Its 3-D figures carry a single repeated caption pairing a CAD model with a tomographic magnitude. Those figures are not reproducible even in principle, which places them alongside the undisclosed sub-aperture count, sub-band overlap and window taper.

8. Robustness: could a wrong velocity model hide a real signal?

The natural objection to a null is that the depth axis is uncalibrated: with no site-specific seismic velocity v(z), might a real reflector be mis-placed and missed? Two versions of this concern must be separated. **Axis calibration is not a threat to the verdict.** A velocity model only maps the relative depth index to metres — a monotonic relabelling of the z-axis, exactly the operation Section 3.2 shows the investigation frequency already performs — and cannot create above-null contrast where none exists. The real tomograms are indistinguishable from their shuffled nulls across the *entire* depth axis, so rescaling that axis leaves the contrast-vs-null unchanged: there is no signal at any depth to relocate.

The legitimate version is **search-grid coverage**: if the inversion's depth grid does not span the depths where a true reflector would sit (because the assumed velocity is wrong), a signal outside the searched range could in principle be missed. The confirmatory test for this is a sweep of the assumed seismic velocity by ±20–50% with the depth grid widened accordingly, re-running the null and positive-control diagnostics at each setting; the null is robust only if no plausible velocity produces an above-null, leakage-clean band. This is the more rigorous form of the velocity check and the one to report in a journal version. The calibration argument above already shows that no relabelling can rescue the current verdict — only a genuine above-null feature could, and there is none to relocate — so the sweep tests grid coverage rather than the conclusion.

9. What is real, and what would change my conclusion

To be fair to the authors: the surface-vibration front-end is legitimate and well-precedented. Their peer-reviewed monitoring of ships, bridges, and the Mosul Dam measures real, millimetric *surface* deformation, and that work stands. My critique is confined to the extrapolation of that surface measurement into deep subsurface structure.

I state my falsifiability explicitly. I would withdraw this conclusion if the method, run with controls on a site of independently documented subsurface structure, produced an above-null, leakage-clean, depth-stable band that matched the known structure in position and depth (after a physically defensible velocity model, not a 22 kHz relabelling). I invite exactly that test. My pipeline, controls, and data are open for it. The cleanest such test would image an intact, independently documented void in the correct mode — for example the Derinkuyu/Cappadocia underground cities — which the free archives do not cover and which would require commercial spotlight tasking; I flag it as the decisive next experiment rather than one the free-data regime permits. Within that regime, Butte (Section 3.3) is the closest available known-target benchmark, and the method does not recover it.

A limitation follows directly from this design. This study evaluates the published method as disclosed in the peer-reviewed papers and the patent (WO2024008365A1), with the fidelity of the reproduction documented block-by-block in Table 1. If unpublished implementation details materially affect the inversion, those details would need to be independently disclosed and evaluated before they could be assessed. The reproducibility framing is therefore deliberate: the conclusion is about the method *as published*, and any essential step absent from the public record is precisely what the original authors would need to provide for the claim to be re-examined.

10. Revision history, corrections and withdrawals

This preprint has been revised five times, and three of those revisions corrected errors in it. That history is set out in full here rather than left for a reader to reconstruct. Disclosed, it is a record of process; discovered, it would be a liability.

10.1 Errors found and corrected

The Komati row (v3, July 2026). The row read "50× / 10×, null" in v2. It had been computed at 128 sub-apertures rather than the default 11 used for every other row, and its verdict label was wrong for that setting. Re-run at the default it gives 2.76 / 1.39. The correction *increases* the margin, because 50/10 is a ratio of exactly 5.0 and sat precisely on this paper's own decision rule. The error was found while stress-testing against an external methodological objection, and was reported to the recommender unprompted. The corrected ratio was then stated as 2.15 from rounded inputs; unrounded it is 1.99.

The 1720× figure (v4, August 2026). Versions 1–3 quoted the Butte reproduction as a "1720× detection" in the abstract, §3.3 and the conclusion. **Withdrawn.** It is a native, unfixed-window peak-to-median contrast computed at 256 sub-apertures; because the depth axis is built as `linspace(0, n_sub·Δz/2, 300)` its extent grows with the sub-aperture count while the bin count stays fixed, so the statistic is not comparable across settings. An earlier "194×" claim had already been withdrawn on exactly this ground and the correction was not carried through. The qualitative result is unchanged.

The rendered-geometry claim (v4). Versions 1–3 concluded that the Giza "shafts", "spirals" and "chambers" *are* the rendered geometry of this artifact. **Withdrawn**, for the reason given in §7: the test of it failed.

The look-shuffle null (v5). Every ratio in Table 2 in versions 1–4 was measured against a null this paper now regards as anti-conservative, in a direction that flattered its own conclusion. See §3.4.

10.2 Explanations proposed and abandoned

Three accounts of the mechanism have been advanced. The first — that the inverter produces the peak from nothing — was asserted before any null existed and was withdrawn once one was built. The second — that 80% sub-aperture overlap manufactures the peak — survived one experiment and died on the next, which found the peak present at *zero* overlap. The third is the account in §5, and it differs from its predecessors in resting on matched controls and a reproduction with the SAR front end removed, rather than on inference from invariance. It also has a failed prediction of its own, at Giza, reported in §5.4. Both abandoned accounts died because the null was built more carefully, which is the same failure this paper identifies in the work it examines.

10.3 Defects in this paper's own analysis code

Three automated verdict messages in the research code printed conclusions their own data did not support: one hard-coded the overlap explanation and never examined the zero-overlap row; one described a peak *bin* as independent of series length when the bins span 4 to 48 and the invariant is the depth in resolution cells; and one announced that removing the cumulative sum moves the peak off the surface, which is false at Giza. All three have been corrected to read the data before printing, and are recorded here because a reader who runs this code should know that its console output was, in three places, written to state the preferred answer.

10.4 What remains open

The constant that fixes the artifact depth at 1.69 rather than the textbook 1.5 cells is not derived. The contrast statistic has not been characterised analytically under strongly autocorrelated inputs; its behaviour is measured across a twelvefold range of series lengths, which is weaker than a derivation. The Giza increments anomaly of §5.4 has no explanation. Two further Giza acquisitions have been obtained and not yet analysed, so the pre-registered within-site repeatability test is untested. The robustness result of §5.5 was obtained on synthetic input only, and no upper bound is offered on how far an adversarially chosen filter could push the contrast ratio; a filter-invariant replacement for criterion (a) is not proposed here and is the most useful thing a reader could contribute. All sites are X-band spotlight; nothing here bears on C-band or L-band. And this preprint has not been peer reviewed.

11. Conclusion

Reproduced from the authors' own method and patent, with each disclosed step mapped to its implementation, single-source SAR Doppler micro-motion tomography yields no reproducible evidence of deep subsurface structure, and the confident outputs it does produce are reproducible as artifacts.

The reported depth in metres is exactly proportional to an investigation frequency chosen by the analyst rather than measured; the frequency never enters the inversion. Across six sites, two independent sensors, eight sub-aperture counts and thirteen patch geometries — 48 runs — the method returns the same surface-pinned feature at 1.2 to 1.9 resolution cells and not one detection. That includes the Giza plateau itself, run against predictions published before the data were processed, and the authors' own Vesuvius. The mechanism is identified: the trajectory is a running total, which has the spectrum of a random walk; the inversion is a Fourier transform, so the reported depth is a frequency; and a degree-2 detrend fixes that frequency near 1.7 cells. Accumulated Gaussian noise with no SAR pipeline at all reproduces both signatures, and at 128 sub-apertures an input containing nothing returns more contrast than a real scene. Planting a signal in the image before processing shows that below 8.4 times the pipeline's own noise a scene containing a real reflector reports *lower* confidence than an empty one.

I do not claim the published three-dimensional imagery is nothing but this artifact; three attempts to reproduce its appearance from empty volumes failed, and that failure is reported in §7. Nor do I offer the contrast ratios here as detection margins: §5.5 shows that statistic can be driven past threshold on empty data by a filter that carries no information, so it is the surface-pinning guard that is doing the work. I do not claim that accumulating displacements is the wrong operation, nor that nothing lies beneath any of these sites. The measurement front end is real and remains valuable for surface-deformation monitoring. What fails reproduction and controls is the deep inference, and the specific defect is located, testable, and reproducible from the repository in minutes.

The 2022 article was retracted by *Remote Sensing* on 10 August 2026 for "serious methodological flaws and statistical errors", without specifying them. I did not contact the journal and make no claim to have prompted that decision. This paper is offered as a reproducibility result — openly testable, falsifiable from the repository, and framed as a critique of method and mathematics, not of intent. I continue to welcome the decisive ground-truth experiment.

Data and code availability

All scenes are free Umbra Open Data and Capella Open Data (CC-BY 4.0); the exact scene identifiers and acquisition parameters are listed in the repository. The full reproduction pipeline, the controls, the stress tests (including the frequency-relabelling and surface-pinning demonstrations), and the scripts that regenerate every figure in this paper are openly available at https://github.com/Hassanforeman/subsurface-sar-tomo (archived at Zenodo, DOI 10.5281/zenodo.21065675), with the code under an MIT licence. Every analysis stage carries a synthetic self-test, so the pipeline can be validated against known truth before it is trusted on real data; a reader can reproduce the nulls, the artifact, the frequency-relabelling, and the stacking result from the scene IDs and the scripts alone.

Conflict of interest disclosure

The author declares that he complies with the PCI rule of having no financial conflicts of interest in relation to the content of the article. The author declares no non-financial conflict of interest.

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Use of artificial intelligence

Artificial-intelligence tools (large language models) were used extensively to assist with software development, data analysis, and the drafting of this manuscript. The author directed the research, made all scientific decisions, and checked and verified every output — code, figures, and text — against the underlying data and the cited sources, and is solely accountable for the content. All figures are generated directly by the analysis pipeline from the source data and are not AI-generated illustrations. No AI system is listed as an author.

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